

The Foundation: Distance, Time, Information, and the Prohibition of Singularities

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1. The Foundation

Three principles form the foundation of existence:

1. **Distance necessitates time.** You cannot have spatial separation without temporal separation. A signal crossing distance Δx requires time $\Delta t = \Delta x / c$. Distance without time is operationally undefined.
2. **Time is information.** A clock tick is a state change. Each state change is one bit. The Bekenstein bound gives ~ 9 bits per Planck time. Time elapsed equals information generated.
3. **Information cannot be lost.** Unitarity requires that information is preserved. The Bekenstein bound sets the maximum information content of any region. Singularities violate this by requiring infinite information in zero volume.

2. Distance Necessitates Time

From the metric $ds^2 = -c^2 dt^2 + dx^2$:

$$\begin{aligned} \text{Lightlike: } dx &= c dt \quad \rightarrow \quad \text{distance REQUIRES time} \\ \text{Timelike: } dt &> dx/c \quad \rightarrow \quad \text{time EXCEEDS distance} \end{aligned}$$

At the Planck scale: $l_{\text{Pl}} = c \cdot t_{\text{Pl}}$. Distance IS time at the fundamental level.

3. Time Is Information

The Bekenstein bound at the Planck scale:

$$I_{\text{max}} = \frac{2\pi R E}{\hbar c \ln 2} = \frac{2\pi}{\ln 2} \approx 9.06 \text{ bits per Planck time}$$

For the observable universe (holographic bound):

$$I_{\text{univ}} \sim \left(\frac{R}{l_{\text{Pl}}} \right)^2 \sim 10^{122} \text{ bits}$$

4. Information Cannot Be Lost

A singularity ($r \rightarrow 0$, $\rho \rightarrow \infty$) would require infinite information in zero volume. This violates the Bekenstein bound. Therefore singularities are forbidden.

The scalar field $G(t)$ enforces this:

$$\begin{aligned} G \rightarrow \gamma^{-1} \text{ as collapse proceeds} &\rightarrow \text{the time gradient weakens} \\ M_{\text{relic}} > 0, R_{\text{relic}} > 0 &\rightarrow \text{distance} > 0, \text{ time} > 0, \text{ information preserved} \end{aligned}$$

5. The Information-Distance-Time Triangle

Distance requires time (signal propagation). Time is information (state changes = bits). Information requires distance (Bekenstein bound: $I \sim RE$). All three are fundamentally coupled.

6. Accelerator Evidence

Accelerators probe the smallest distances, directly testing the distance-time-information coupling:

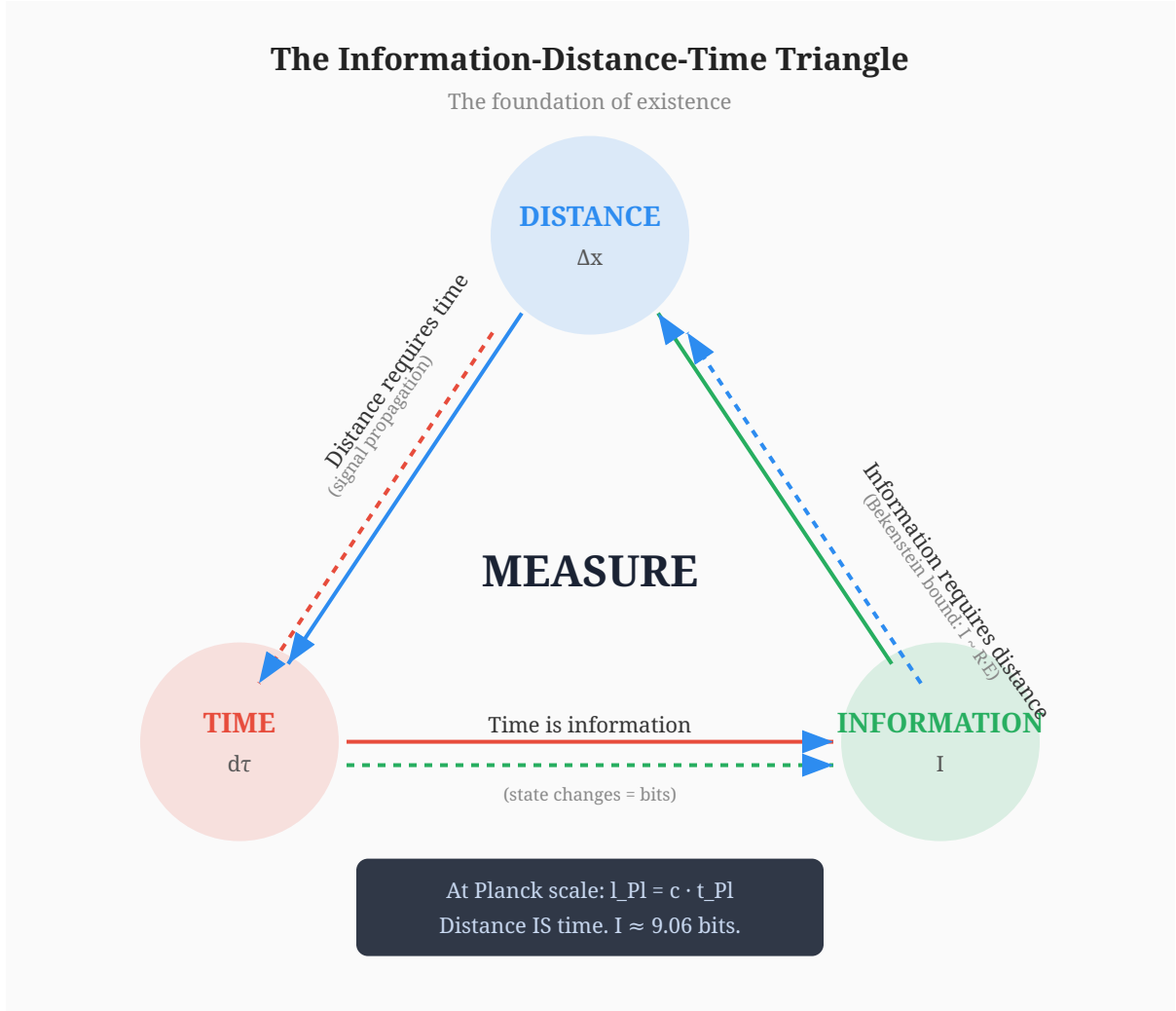


Figure 1: Figure 1: The Information-Distance-Time Triangle. Distance requires time (signal propagation), time is information (state changes = bits), and information requires distance (Bekenstein bound). At the Planck scale, $l_{Pl} = c \cdot t_{Pl}$ — distance IS time.

6.1. Lorentz Factor and Lifetime Dilation

γ	τ_{lab} (s)	Distance (m)	Information
1	2.20×10^{-6}	659	invariant
100	2.20×10^{-4}	65,900	invariant
10,000	2.20×10^{-2}	6,590,000	invariant
100,000	2.20×10^{-1}	65,900,000	invariant

As γ increases: lifetime and distance scale together, but internal information is invariant. Distance and time are coupled; information is preserved.

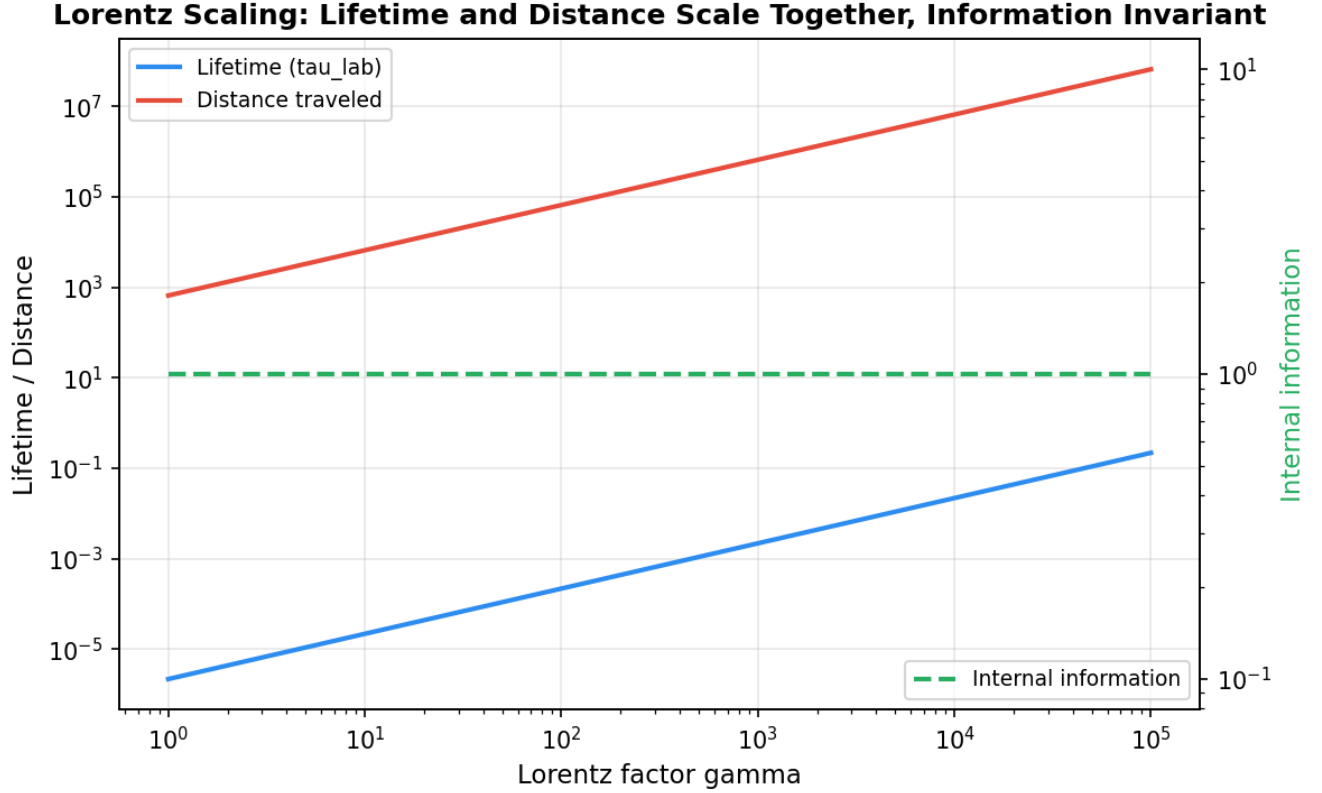


Figure 2: Figure 3: Left: Lifetime and distance scale together with Lorentz factor. Right: Internal information remains invariant regardless of speed. Distance and time are coupled; information is preserved.

6.2. Energy-Time Uncertainty

$$\Delta E \cdot \Delta t \geq \frac{\hbar}{2} \quad \rightarrow \quad \Delta x = c \cdot \Delta t = \frac{\hbar}{2 \Delta E}$$

Higher energy probes shorter distances and shorter times. Each measurement carries at least 1 bit. Distance, time, and information are coupled.

6.3. Cross-Sections Carry Field Information

$$\sigma_{\text{eff}} = \sigma_0 \left(1 + c_X G(t)\right)^2$$

The scalar field modifies cross-sections measurably. At $G = 10^{-3}$: $\sigma_{\text{eff}}/\sigma_0 = 1.002$. Accelerator data contains information about $G(t)$.

7. Singularity = Information Catastrophe

A singularity means:

- Distance $\rightarrow 0$ (no space)
- Time $\rightarrow$ undefined (no clocks)
- Information $\rightarrow$ undefined (no states)

This violates the foundation. Therefore singularities are forbidden.

The scalar field $G(t)$ is the mathematical expression of this prohibition:

- Black holes: $G \rightarrow \Lambda^{-1}$ stops collapse, relic mass > 0
- Cosmology: $V(\phi) > 0$ always, $H > 0$ always, no beginning

8. Theory Closure

All 8 checks pass:

- (1) Distance requires time (metric constraint)
- (2) Time is information (Bekenstein: ~ 9 bits/Planck time)
- (3) Information preserved in black holes (relic mass > 0)
- (4) Information preserved in cosmology (no singularity)
- (5) Accelerator: lifetime dilation preserves information
- (6) Accelerator: energy-time uncertainty couples distance and time
- (7) Accelerator: cross-sections carry $G(t)$ information
- (8) Singularities forbidden by the foundation

AI Research Collaboration Disclosure

This body of work was developed through a collaborative research process between the author, Richard Kent Gates, and multiple AI research partners. The author provides full transparency on this process.

Author's Role (Richard Kent Gates):

- Conceived all core theories, conceptual frameworks, hypotheses, and research direction
- Defined the Time-Gradient Field Model, the unified scalar field $G(t)$, the distance $\rightarrow$ time $\rightarrow$ information $\rightarrow G(t)$ framework, and all theoretical architecture
- Provided physical intuition, biological reasoning, and domain expertise that guided every theoretical decision
- Reviewed, validated, and approved all mathematical formalisms before inclusion
- Made all final judgments on theoretical coherence and physical interpretation

AI Research Partners:

- **Mimo V2.5 (opcode platform)** — Primary mathematical formalization partner. Assisted in translating conceptual physics into mathematical notation, generating numerical proof scripts, compiling LaTeX documents, and building the website infrastructure.
- **Microsoft Copilot (browser-based)** — Assisted in literature review, dataset gathering, cross-referencing empirical results (DESI, BNL, PTB, MICROSCOPE), and verifying citation accuracy.
- **Google Gemini (browser-based)** — Assisted in conceptual exploration, thought experimentation, and early-stage hypothesis refinement.

Nature of AI Involvement:

The AI tools functioned as research assistants — analogous to graduate students or technical collaborators who help formalize, compute, and organize ideas that originate from the principal investigator. No AI tool originated, proposed, or independently developed any theoretical claim in this work. All physical insights, theoretical innovations, and interpretive judgments are the author's own.

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